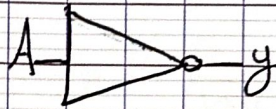


Ch:2 Algebre de boole.

V $\rightarrow$ 1
F $\rightarrow$ 0

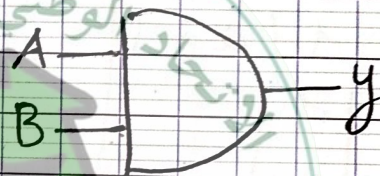
1) Table de vérité et porte logique de Non (NOT):

A	$\bar{A}$
0	1
1	0



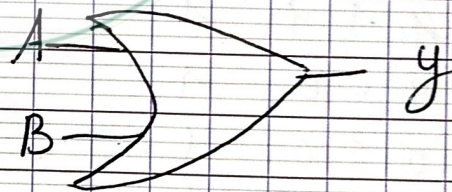
2) Table de vérité et porte logique de ET (AND):

A	B	$A \cdot B$
0	0	0
1	0	0
0	1	0
1	1	1

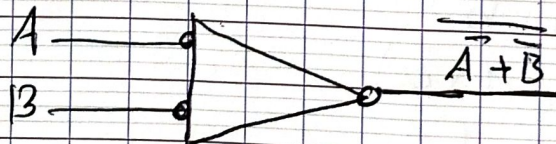


3) Table de vérité et porte logique de OU (OR):

A	B	$A + B$
1	1	1
1	0	1
0	1	1
0	0	0



$$A \cdot B = \overline{\overline{A \cdot B}} = \overline{\overline{A} + \overline{B}}$$





propriétés des
opérateurs AND
et OR

$$a + 0 = a$$

a	0	a+0
1	0	1
0	0	0

$$a = a + 0$$

a	1	a.1
1	1	1
0	1	0

a	a	a+a
1	1	1
0	0	0

a	$\bar{a}$	$\bar{a} + a$
1	0	1
0	1	1

a	1	a+1
0	1	1
0	0	0

$$a + a \cdot b = a(1 + b) = 0$$

a	b	ab	$a + b$	$a(1+b)$	$a + ab$
1	1	1	1	1	1
1	0	0	1	1	1
0	1	0	1	0	0
0	0	0	1	0	0

$$(a + b) \cdot b = ab$$

$$ab + b\bar{b} = ab + 0 = ab$$

$$\bar{a}b + b = a + b$$

$$a\bar{b} + b(1+a)$$

$$a\bar{b} + b \cdot 1$$

$$= a\bar{b} + b(1+a)$$

$$= a\bar{b} + b + ab$$

$$= a(\bar{b} + b) + b$$

$$= a \cdot 1 + b = a + b$$



$$A \cup B = \{x \in E, x \in A \text{ or } x \in B\}$$
$$\longrightarrow A + B$$

$$A \cap B = \{x \in A \text{ and } x \in B\}$$

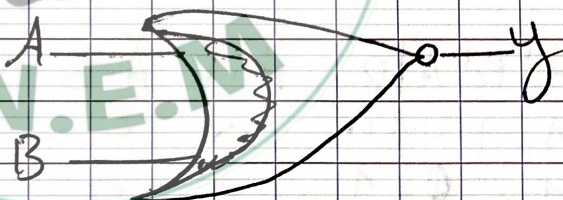
$$A \cap B \longrightarrow A \cdot B$$

$$C_A^E = \{x \in E, x \notin A\}$$
$$\longrightarrow \bar{A}$$

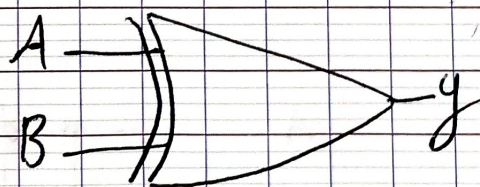
A	B	$\bar{A} \cdot B$
0	0	1
1	0	1
0	1	1
1	1	0



A	B	$A + B$
0	0	1
1	0	0
0	1	0
1	1	0

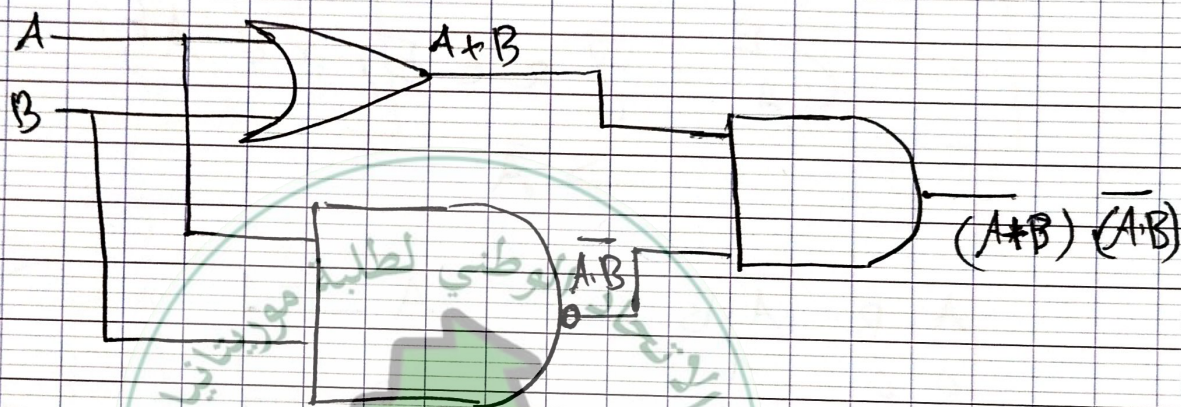


A	B	$A \oplus B$
1	0	1
0	0	0
1	1	0
0	1	1



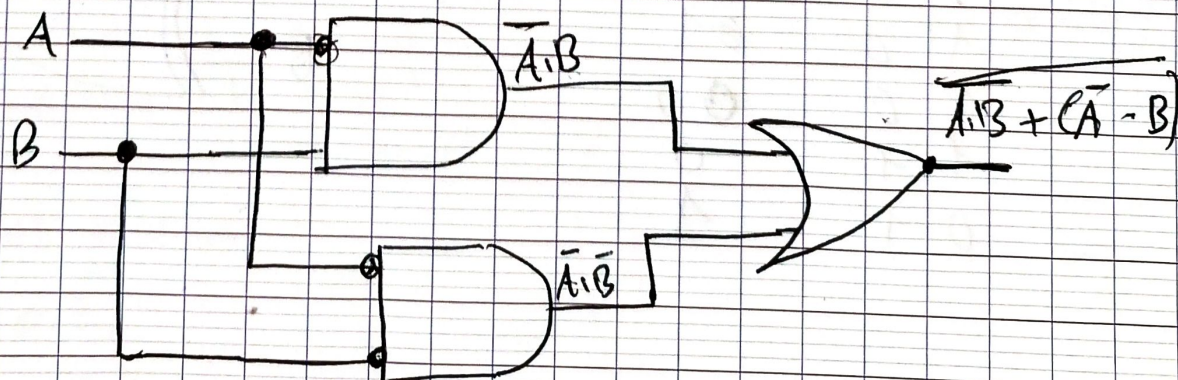
$$A \oplus B = (A+B) - (\overline{A \cdot B})$$

A	B	A+B	A·B	$\overline{A \cdot B}$	$(A+B) - (\overline{A \cdot B})$	$A \oplus B$
1	1	1	1	0	0	0
1	0	1	0	1	1	1
0	1	1	0	1	1	1
0	0	0	0	1	0	0



$$A \oplus B = (A \cdot \overline{B}) + (\overline{A} \cdot B)$$

A	B	$\overline{A}$	$\overline{B}$	$A \cdot \overline{B}$	$\overline{A} \cdot B$	$(A \cdot \overline{B}) + (\overline{A} \cdot B)$	$A \oplus B$
1	1	0	0	0	0	0	0
1	0	0	1	1	0	1	1
0	1	1	0	0	1	1	1
0	0	1	1	0	0	0	0



$$\begin{aligned}
 F_3(a, b, c) &= \overline{a} \overline{b} + \overline{a} b c \\
 &= \overline{a} \overline{b} \cdot 1 + \overline{a} b \cdot c \\
 &= \overline{a} \overline{b} \cdot (c + \overline{c}) + \overline{a} b \cdot c \\
 &= \overline{a} \overline{b} c + \overline{a} \overline{b} \overline{c} + \overline{a} b c
 \end{aligned}$$